Sadeeq Adeyemi

Perception Engineer, 3D Computer Vision and Multi-Object Tracking

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SUMMARY

Perception engineer in 3D computer vision and deep learning. I build trackers that keep an object's identity while it is hidden, and prove them on public benchmarks and real driving data. Shipping machine learning in industry since 2018, from a vision model on a steel plant floor to research on Meta's SAM 3.

SKILLS

Tracking and estimation: multi-object tracking and segmentation (MOTS), Kalman filters (EKF, UKF), data association (Hungarian, JPDA, MHT), sensor fusion

3D vision and deep learning: lidar point clouds, camera and lidar projection, ray casting, instance segmentation, SAM 3, YOLOv8, re-identification

Software: Python, C++17, PyTorch, ROS 2, OpenCV, filterpy, MATLAB, Git, Docker, Linux, AWS, AI coding agents

Evaluation: HOTA, IDF1 and MOTA with TrackEval, significance testing; KITTI, KITTI-MOTS, MOTS20, MOT17, NVIDIA PhysicalAI-AV

EXPERIENCE

Graduate Research Assistant, CVAR Lab (Computer Vision and Robotics), University of Maine 2025 – present

- Benchmarked Meta's SAM 3 and SAM 3.1 on KITTI-MOTS and MOTS20 with the official HOTA evaluation on cloud A100 GPUs. SAM 3 scored 82.6 car HOTA (val) and kept identity through 94% of 245 real occlusions; SAM 3.1 scored 74.6 and 63%.
- Traced about half of SAM 3.1's deficit to its shipped tracking settings by reading Meta's source code; a controlled rerun raised its car HOTA from 74.0 to 78.3. Isolated a separate fault that halts it on long, crowded sequences.
- Built a Kalman-filter identity layer on SAM's output that returns a lost ID when the object reappears, raising SAM 3.1's identity kept through occlusion from 63% to 70% ($p = 0.004$).
- Built a pipeline that mined 10,871 pedestrian occlusions from 1,256 NVIDIA driving clips in 12 countries, using 3D ray casting against labelled boxes and lidar depth projected into a fisheye camera.
- Showed on that real street geometry that an occlusion-aware tracker keeps identity through 1 to 5 second hides in 57% of 1,600 cases, where a standard tracker keeps 2%; driven by a real YOLOv8 detector, 44% against 3%.
- Built mobile robots end to end, from sensors and wiring to ROS 2 nodes; implemented EKF localization fusing IMU, wheel odometry and lidar, with costmap planning and obstacle avoidance.

Materials Engineer, R.W. Gillespie, Maine 2024

- Ran cone penetration tests (CPT) and materials testing for construction projects in Maine.

Machine Learning Engineer and Structural Steel Fabricator, NewTech Steel, Nigeria 2018 – 2023

- Built and deployed a computer vision model that detects fatigue and defects in steel in real time, halving inspection time by automating a manual test.
- Led fabrication and erection of structural steel for factory and auditorium projects.

Civil Engineering Intern, Federal Capital Development Authority, Abuja, Nigeria Feb – Aug 2017

- Worked on the construction of wastewater treatment facilities for Nigeria's Federal Capital Territory.

EDUCATION

MS, Data Science and Engineering, University of Maine 2025 – present

MS, Structural Engineering, University of Maine 2023 – 2025

MSc, Project Management, University of Portsmouth, United Kingdom 2023

BEng, Civil Engineering, Federal University of Technology Akure, Nigeria 2019

Certifications: AWS Certified Machine Learning Specialty; AWS Certified Solutions Architect, Associate